Searching for counterexamples to Gale's conjecture*

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September 2026

1 Introduction

This paper describes algorithms for finding finite point sets whose partitions into a prescribed number of parts necessarily contain a part of large diameter. The approach combines graph-coloring queries with continuous optimization of point coordinates, and extends this search by modifying both the witness graph and the number of points. It accompanies *A counterexample to Gale's conjecture in* $\mathbb{R}^3$, which presents the geometric constructions and their consequences for the quantitative Borsuk problem. Here the emphasis is on the computational methods: how promising configurations are generated, refined, and checked, and how discrete and continuous steps are combined in a reproducible search.

We write $R_k(X)$ for the diameter ratio of a finite configuration X, denoted by $\beta_k(X)$ in the companion paper; its formal definition is recalled in Section 2.1. The computational objective is to maximize this ratio over the coordinates. Evaluating a single configuration already requires a discrete optimization: one must determine how its points can be assigned to k parts. Moreover, the relevant distance constraints change as the points move. The resulting objective is non-smooth, and a local coordinate optimization can stagnate even when changing the underlying combinatorial constraints would permit further improvement.

The central device is a non-k-colorable *witness graph* on the point indices. Its shortest edge, divided by the diameter of the configuration, gives a lower bound for $R_k(X)$. Holding this graph fixed therefore turns the geometric search into a continuous problem with explicit distance constraints. Graph-coloring queries enter at two complementary stages: threshold graphs evaluate the current configuration, while witness pruning removes constraints that are unnecessary for the retained non-colorability obstruction. Alternating these steps allows the search to change the obstruction it optimizes, rather than remaining tied to the first witness found at initialization.

The baseline method combines multiple initializations and geometric perturbations with local optimization of a fixed witness. A continuation of smooth

*This AI-written document complements the article https://arxiv.org/abs/2610.28828

approximations is optimized by L-BFGS-B, followed by constrained polishing with SLSQP. The tuning procedures add witness refresh, vertex deletion, and point insertion, together with a linear-programming diagnostic of possible first-order improvement. These operations serve different purposes: coordinate motion improves a realization of a graph, pruning changes its active requirements, and insertion or deletion changes the geometric search space. Their interaction is therefore described explicitly, including the distinction between exploratory states and the configuration retained as the incumbent.

Vertex splitting provides a further way to change the constraints acting on a configuration. A point is replaced initially by two coincident copies whose incident witness edges differ, while the resulting graph remains non-k-colorable. Subsequent optimization can separate the copies and exploit motions unavailable to the original vertex. We describe both single-vertex splitting and a paired procedure that constructs two successive splits before optimizing their joint realization. The latter explores the possibility that a first split enables a useful second move without itself yielding an immediate geometric improvement. Its current implementation is restricted to $k = 4$. Coincident copies are auxiliary realizations during proposal construction; saved finite-set certificates concern distinct points.

A common requirement throughout these methods is to separate candidate generation from the establishment of bounds. Numerical losses, predicted LP gains, and optimizer termination reports guide the search. Lower bounds are instead obtained from non-colorable witnesses, and upper bounds from explicit partitions. For saved decimal coordinates, the corresponding squared-distance comparisons are performed with integers and rational numbers. Unresolved coloring queries retain a bracket and are never treated as proofs of non-colorability. The graph solver's non-colorability result remains part of the trusted computation: the implementation does not provide independent SAT proof checking. Thus the search is heuristic, while its reported finite-set bounds have a separate, explicit verification procedure.

The algorithmic account proceeds from the baseline evaluator and search in Sections 2.1–2.10 to exact-decimal tuning in Sections 2.11–2.20, single-vertex splitting in Sections 2.21–2.27, and paired splitting in Sections 2.28–2.33. The description includes proposal screening, acceptance rules, stopping conditions, and the scope of the saved certificates. These details are necessary for interpreting a computational improvement: the workflows use different incumbent policies, and neither successful local optimization nor exhaustion of a search budget establishes global extremality.

2 Algorithms

The first part of this section describes the baseline search implemented in `borsuk_search.py`. The method alternates a discrete graph-coloring calculation with continuous optimization of a non-colorable witness graph. Independent restarts and geometric perturbations explore different local solutions. Sec-

tions 2.11–2.20 describe the post-processing algorithms of `tune_solution.py`. Sections 2.21–2.27 develop the single-vertex splitting method of `split_tune.py`. Sections 2.28–2.33 describe paired splitting in `paired_splits.py`. The continuous steps are heuristic; bounds are computed separately from witness graphs and explicit partitions. The dimension d, number of parts k, and number of points m are parameters, with $k = d + 1$ by default.

2.1 Objective and graph formulation

For $X = \{x_1, \dots, x_m\} \subset \mathbb{R}^d$ of diameter $D > 0$, define

$$R_k(X) = \min_{c:\{1,\dots,m\}\to\{1,\dots,k\}} \frac{\max_{i<j:\, c(i)=c(j)} \|x_i - x_j\|}{D}.$$

The maximum over an empty set is zero, so empty parts and singletons are allowed. The search seeks large values of $R_k(X)$ at fixed d and k; m is fixed within a restart. For $m \le k$, the value is zero.

For $t \ge 0$, let $G_t(X)$ contain the edge $\{i, j\}$ precisely when $\|x_i - x_j\| > tD$. A proper k-coloring of this graph is equivalent to a partition with maximum part diameter at most tD. Hence

$$G_t(X) \text{ is } k\text{-colorable} \quad \Longleftrightarrow \quad R_k(X) \le t.$$

More generally, any graph H on the point indices known to be non-k-colorable gives the lower bound

$$L_H(X) = \frac{\min_{\{i,j\}\in E(H)} \|x_i - x_j\|}{D} \le R_k(X). \tag{1}$$

Indeed, every assignment of k colors has a monochromatic edge of H. Conversely, any explicit assignment c gives an upper bound $U_c(X)$ by its maximum within-part distance divided by D. The witness need not contain every edge of a geometric threshold graph.

2.2 Initialization and normalization

Each restart receives a random seed from a child of a common seed sequence. When a point-count interval is supplied, m is sampled uniformly from its integer values. The mixed initialization option cycles through six families:

1. `simplex-edges`: samples on edges of a regular d-simplex, radially projected to the unit sphere;
2. `prism-edges`: samples on the edges of a prism whose two bases are regular $(d-1)$-simplices;
3. `sphere`: independent Gaussian vectors normalized individually, giving uniform samples on the unit sphere;

4. `simplex-chords`: the simplex-edge construction without radial projection;

5. `prism-sphere`: the prism-edge construction with radial projection;

6. `simplex-faces`: convex combinations of three randomly selected simplex vertices, followed by radial projection.

The simplex vertices have unit norm and pairwise inner product $-1/d$. Edge sampling includes the vertices when the point budget permits and allocates the remaining points approximately evenly among edges. For an edge receiving q samples, the interpolation parameters are $(j+1/2+\xi_j)/q$, where $j = 0, \dots, q-1$ and $\xi_j \sim \mathcal{U}[-0.3, 0.3]$. If fewer points than vertices are requested, a random subset of vertices is used. Prism bases lie at heights $\pm h$, where $\log h$ is uniform on $[\log 0.3, \log 1.2]$. The face initializer retains up to $d+1$ simplex vertices and draws the remaining barycentric weights from Dirichlet$(0.65, 0.65, 0.65)$.

Except for the uniform-sphere family, seed coordinates receive Gaussian jitter of standard deviation $\eta/\sqrt{d}$ per coordinate, with default $\eta = 0.015$, and are normalized again. Free-coordinate search subtracts the centroid and divides by the actual diameter. An optional sphere constraint instead normalizes every row to unit norm throughout the optimization. A spherical initializer alone does not impose this constraint on subsequent search. An existing configuration may also be supplied: the first restart uses its normalized coordinates, and later restarts apply a mutation before evaluation.

2.3 Computation of the edge list and finite-set evaluation

All $N = \binom{m}{2}$ unordered pairs are indexed once. For each new geometry, the program computes $s_{ij} = \|x_i - x_j\|^2$ and stably sorts the pairs into an order $e_1, \dots, e_N$ with $s_{e_1} \geq \cdots \geq s_{e_N}$. This takes $O(m^2 d)$ arithmetic operations for the distances and $O(m^2 \log m)$ for sorting. Let P_q contain the first q edges. Colorability is monotone along this sequence: adding edges cannot restore colorability. When $m > k$, P_0 is colorable and $P_N = K_m$ is not.

The evaluator maintains a colorable prefix length ℓ and a non-colorable prefix length h, initially 0 and N. It queries $P_{\lfloor(\ell+h)/2\rfloor}$. A colorable answer increases ℓ; a non-colorable answer decreases h. A returned coloring can additionally advance ℓ to include every edge strictly longer than its maximum monochromatic distance. If an assumption core is returned with a non-colorable answer, h can be reduced to the largest edge rank in that core. At most $O(\log N)$ oracle calls are needed to resolve the transition, although a coloring call itself can have exponential cost.

At any unresolved stage, the prefix bounds are

$$\sqrt{\frac{s_{e_h}}{s_{e_1}}} \leq R_k(X) \leq \sqrt{\frac{s_{e_{\ell+1}}}{s_{e_1}}}, \tag{2}$$

with the upper endpoint tightened by a retained coloring when available. If q is the first non-colorable prefix length, then $R_k(X) = \sqrt{s_{e_q}/s_{e_1}}$. In zero-based

array notation this uses `distance[q-1]`, not `distance[q]`. Tied distances are kept unchanged: a prefix inside a tied block need not itself be a strict threshold graph, but the bounds and transition formula remain valid. The floating calculation is marked resolved when the two prefix lengths are consecutive or the computed endpoints coincide. This is a statement about numerical evaluation; saved decimal geometry is treated separately in Section 2.10.

2.4 Graph-coloring oracle

The oracle distinguishes SAT (colorable), UNSAT (non-colorable), and UNKNOWN. An unknown answer terminates the current threshold search with its existing bracket. A timeout or solver failure does not establish non-colorability.

The SAT encoding uses Boolean variables b_{ic} for assigning color c to vertex i. Each vertex receives exactly one color through one positive clause and pairwise exclusion clauses. Every graph edge adds $\neg b_{ic} \vee \neg b_{jc}$ for each color. Color-permutation symmetry is reduced by requiring colors to appear in order of first occurrence: using color $c > 1$ at vertex i requires an earlier occurrence of color $c - 1$. Every coloring admits such a relabeling.

External DIMACS solvers receive a fresh formula for each uncached query. The incremental PySAT option instead attaches an activation literal to each potential edge and queries a graph by assuming its edge literals. It reuses learned clauses and may return an unsatisfiable assumption core. The other options are a native DSATUR library and a Python DSATUR fallback. The Python implementation first peels vertices of degree less than k, then branches on an uncolored vertex of maximum saturation (number of neighbor colors), breaking ties by remaining degree. It tries existing colors and at most one new color to remove color-permutation symmetry. After a successful search, peeled vertices are colored in reverse order. Timeouts and recursion limits return UNKNOWN.

Resolved graph answers are cached by their edge sets, with at most 2048 entries; unknown answers are not cached. Returned coloring arrays are checked for length, color range, and edge conflicts. An external SAT status may arrive without a usable model: it supplies no explicit partition for export. Automatic backend selection prefers an available external solver, then PySAT, then the native library, then Python.

2.5 Witness enlargement and edge pruning

At each round, let a be the shortest edge length in the current known non-colorable witness. The program forms the enlarged graph

$$E = \{\{i, j\} : s_{ij} \geq a^2(1 - \delta)^2\}.$$

Here $\delta = 0$ in the first round and subsequently is uniform on $[\sigma/4, \sigma]$, where the default graph slack is $\sigma = 0.025$. This graph contains the old witness and therefore remains non-colorable. Enlargement permits the pruning stage to select a different obstruction, including edges slightly shorter than the previous bottleneck.

Pruning first repeatedly removes edges incident to vertices of degree less than k. This k-core reduction preserves non-k-colorability: a coloring of the remaining graph would extend to each removed vertex in reverse order. If the solver returns a smaller assumption core, that core is peeled as well. Remaining edges are ordered by

$$\frac{s_e}{D^2} + \nu Z_e, \qquad Z_e \sim \mathcal{N}(0,1),$$

so shorter edges are preferentially tested for deletion, with randomized tie and priority changes. During search rounds, $\nu = 0.015 + \delta$. For each tested edge, deletion is accepted only when the remaining graph receives a definite UNSAT answer. A returned core can further reduce the graph, followed by another peeling step. Both SAT and UNKNOWN retain the edge. The default search budget is 120 trial deletions per pruning call, in addition to its initial query. This bounded greedy procedure need not produce an edge-minimal witness, and does not solve a minimum-cardinality problem. Peeling removes graph constraints, not coordinate rows.

2.6 Smooth optimization of a fixed witness

For a fixed non-colorable witness H, the continuous step seeks to increase $L_H(X)$. With positive squared distances, define

$$F_p(X) = \frac{1}{p}\log \sum_{e \in \binom{[m]}{2}} s_e^p + \frac{1}{p}\log \sum_{e \in E(H)} s_e^{-p}, \qquad p > 0. \tag{3}$$

Writing $A = \max_e s_e$ and $B = \min_{e \in E(H)} s_e$, one has

$$\log(A/B) \le F_p(X) \le \log(A/B) + \frac{\log N + \log |E(H)|}{p}.$$

Thus minimizing this scale-invariant loss increasingly approximates maximizing the witness ratio as p grows. The implementation evaluates the sums by log-sum-exp and clips squared distances below 10^{-24} before taking logarithms. At clipped entries the derivative is zero; the approximation and scale invariance above apply away from clipping.

Analytic gradients follow from weights

$$\alpha_e = \frac{s_e^p}{\sum_f s_f^p}, \qquad \beta_e = \frac{s_e^{-p}}{\sum_{f \in E(H)} s_f^{-p}} \quad (e \in E(H)),$$

with $\beta_e = 0$ outside the witness. Edge $\{i, j\}$ contributes $2(\alpha_e - \beta_e)(x_i - x_j)/s_e$ to the gradient at i and its negative at j. In sphere mode the optimization variables y_i are mapped to $x_i = y_i/\|y_i\|$, and the gradient is multiplied by $(I - x_i x_i^\top)/\|y_i\|$, with a numerical radius floor.

The optimizer applies L-BFGS-B successively at $p = 8, 24, 80, 256, 1024$, using each stage's endpoint to initialize the next. The default limit is 180 iterations

per stage. Coordinates are normalized between stages. Independently of optimizer termination, every finite objective evaluation is considered for retention using the actual unsmoothed witness ratio. The best such geometry, including the initial geometry, is returned. A lower smooth loss alone is not an acceptance criterion for a scientific result.

2.7 Relocation of inactive points and basin hopping

In free-coordinate mode, points incident to no edge of the optimized witness are relocated. Each is placed at $(1-t)x_a + tx_b$ for two randomly selected distinct active vertices and $t \sim \mathcal{U}[0.04, 0.35]$. The active points and witness lengths remain unchanged, while placing all other points in their convex hull cannot increase the diameter. The witness lower bound therefore cannot decrease. The next discrete evaluation recomputes all distances and may involve these points in a new witness. This relocation is skipped in sphere mode.

The newly evaluated geometry becomes the current exploratory state even if it does not improve the saved lower endpoint. After two consecutive rounds without a saved improvement, the algorithm mutates the restart's best geometry. With equal probabilities, it perturbs a random subset of vertices (selection probability $1/4$, forcing at least one selection), applies a random linear map $I + \mu A$ with independent standard Gaussian entries in A, or perturbs every coordinate with Gaussian standard deviation $\mu/\sqrt{d}$. Subset perturbations have standard deviation μ per coordinate. For these stagnation-triggered mutations, μ is uniform on $[0.4\mu_0, 2\mu_0]$, where $\mu_0 = 0.035$ by default. The mutated configuration is normalized and evaluated anew; it never inherits a bound computed for its parent coordinates.

2.8 Application of SLSQP

At the end of a normally completed free-coordinate restart, the best configuration's witness is pruned again and polished using sequential least-squares quadratic programming (SLSQP). With coordinates initially normalized to diameter one and an auxiliary variable t, the problem is

$$\begin{aligned} \text{maximize} \quad & t, \\ \text{subject to} \quad & \|x_i - x_j\|^2 \le 1 && (i < j), \\ & \|x_i - x_j\|^2 \ge t && (\{i,j\} \in E(H)). \end{aligned} \tag{4}$$

All pairwise diameter constraints are included. Analytic constraint Jacobians are supplied, coordinates are bounded to $[-2, 2]$, and $t \in [0, 1]$. The default limit is 100 iterations and the numerical termination tolerance is 10^{-12}. Sphere-constrained restarts skip this free-coordinate polish.

The routine retains the best actual witness ratio among its initial geometry, callback iterates, and final iterate. In particular, neither numerical feasibility nor the reported value of t is used as a certificate: the diameter and minimum witness length are recomputed, even after an unsuccessful solve. The retained

coordinates are normalized and then subjected to the full discrete evaluator. SLSQP provides local proposals and does not establish global optimality or exact stationarity.

2.9 Multistart orchestration and stopping

A restart can be summarized as follows:

1. Initialize and normalize the coordinates; evaluate and save the initial bracket.
2. Enlarge and prune a known non-colorable witness.
3. Optimize the fixed witness by the continuation of smooth losses; relocate inactive points when permitted.
4. Reevaluate the complete finite-set objective and update the saved candidate if its lower endpoint improves.
5. Check optional early stopping; otherwise mutate after stagnation and repeat up to the round budget.
6. Unless early-stopped, apply optional SLSQP polishing, reevaluate, and retain an improvement if found.

The default budget is 12 restarts with 10 rounds each. Restarts can run in separate worker processes, each with its own coordinates and oracle. Within a restart, a new floating lower endpoint must exceed the saved one by more than 10^{-13}. The global output is selected by the largest returned restart lower endpoint. These rules rank lower bounds; they do not require a new lower bound to exceed an earlier upper bound. With unresolved brackets, the label "best" therefore does not prove an ordering of the exact finite-set objectives.

Optional early stopping shares the largest saved, outward-rounded lower endpoint among workers. Let this record be L_* and let U_{seen} be the largest evaluation upper endpoint observed in the current restart. At an active round checkpoint with allowed gap g, the restart stops if

$$U_{\text{seen}} + 10^{-12} < L_* - g.$$

Using the largest observed upper bound avoids interpreting a weak lower bound caused by a timeout as evidence of poor quality. The optional default schedule uses $g = 0.02$ from round 2 and $g = 0.005$ from round 5, with checks after each subsequent round. The first restart and every tenth restart are protected by default. Early stopping skips remaining rounds and final polishing. It is a throughput heuristic, not a bound on future improvement. Parallel scheduling can affect stopping decisions even when the random seeds are fixed.

2.10 Exact-decimal export and verification

Saved configurations are rounded to 17 decimal places and written as rational coordinates $x_i = z_i/10^{17}$ with integer vectors z_i. If rounded rows coincide, the first coordinate of each repeated row is incremented in units of 10^{-17} until it is distinct; the number of adjusted rows is recorded. This explicitly changes the saved set, so its geometry is recomputed. The same abstract non-colorable graph remains a witness after rounding or separation, although its geometric bound may change. Any retained color assignment likewise remains an admissible partition, with a newly computed upper bound.

Using integer squared distances $q_{ij} = \|z_i - z_j\|^2$, the exporter computes

$$A = \max_{i<j} q_{ij}, \qquad B = \min_{\{i,j\}\in E(H)} q_{ij}, \qquad C = \max_{i<j:\, c(i)=c(j)} q_{ij}.$$

The saved bounds are $\sqrt{B/A} \le R_k(X) \le \sqrt{C/A}$. An empty witness gives $B = 0$; without an explicit partition the exporter uses $C = A$. Integer square roots produce downward and upward rounded decimal endpoints, respectively, at 12 decimal places. Equal rational squared endpoints determine the objective of this saved finite set, conditional on graph non-colorability; floating threshold resolution alone does not assert this equality after export.

Each snapshot contains the authoritative decimal CSV coordinates, a binary NumPy convenience copy, witness edges and their DIMACS formula, a partition when available, and certificate metadata. Verification reads the CSV as exact rationals, checks dimensions, distinctness and partition labels, reruns the witness-coloring query, and recomputes the integer geometry and stated bounds. The NumPy file is not assumed to represent the identical rational set. Non-colorability relies on the solver or complete graph base case; the implementation does not include an independent SAT proof checker. These computations establish finite-set bounds and do not certify global extremality over configurations.

2.11 Exact-decimal tuning: state and initialization

The post-processing program `tune_solution.py` starts from an existing configuration and combines vertex deletion, alternative witness selection, point insertion, and repeated local optimization. It uses the graph oracle and smooth witness optimizer described above, but maintains exact rational squared bounds throughout discrete evaluation and incumbent comparison. Its command-line workflow requires $k \ge 2$ and $m > k$.

A geometry consists of distinct integer rows $z_i \in \mathbb{Z}^d$ and a common decimal precision p, representing $x_i = 10^{-p} z_i$. Decimal CSV input is read without changing its coordinates: p is chosen to accommodate all entries, and conversion to integers uses exact rational arithmetic. Non-CSV input is loaded numerically, normalized, and rounded to 17 decimal places. Subsequent continuous proposals likewise produce new geometries at 17 decimal places. Unlike the baseline exporter, the tuner rejects coincident decimal rows rather than separating them.

A floating copy is used for numerical optimization, while all pairwise squared distances

$$q_{ij} = \|z_i - z_j\|^2, \qquad A = \max_{i<j} q_{ij}$$

are stored as integers. Subsetting retains the original integer rows and recomputes A.

A state consists of this geometry, a non-colorable witness H, an explicit partition c, and rational squared endpoints

$$\lambda = \frac{\min_{e \in E(H)} q_e}{A}, \qquad \upsilon = \frac{\max_{i<j:\, c(i)=c(j)} q_{ij}}{A}.$$

Thus $\sqrt{\lambda} \le R_k(X) \le \sqrt{\upsilon}$. Equality is tested as equality of rational numbers, not within a floating tolerance. An unresolved-query flag and numerical polishing diagnostics are kept separately. Saved witness and partition files can seed the initial assessment; the witness is rechecked. Changing the requested k discards these inherited graph data. A pool retains up to six graph oracles indexed by point count, permitting solver reuse as deletion and insertion change m.

2.12 Integer edge levels and seeded evaluation

The tuner replaces the floating prefix search of Section 2.3 with a search over distinct integer squared distance levels. Initially, the complete graph gives the lower numerator $b = \min q_{ij}$, and the supplied partition gives an upper numerator $u = \max_{c(i)=c(j)} q_{ij}$. Without a supplied partition, vertices are assigned cyclically to the k colors. A verified supplied witness replaces the initial complete-graph witness. Internal callers may reuse a graph whose non-colorability was already established, since changing its geometric realization does not change its chromatic number.

While $b < u$, select a middle distinct distance value v in $(b, u]$ and query the inclusive graph

$$E_v = \{\{i, j\} : q_{ij} \ge v\}.$$

A SAT answer with a model replaces the partition and sets u to its actual maximum monochromatic squared distance, which is strictly less than v. An UNSAT answer replaces the witness by the returned assumption core, when available, or by E_v, and sets b to its shortest squared edge length, at least v. Entire tied levels are included, ensuring progress without a tie-breaking perturbation. A query returning UNKNOWN, or SAT without a usable model, stops this loop with the established bracket. An unresolved supplied-witness check can instead leave the complete-graph lower bound in place while threshold evaluation proceeds. The final endpoints are recomputed from the retained witness and partition.

A normalized threshold $\rho = a/b$ for squared ratios is also constructed without division: edge $\{i, j\}$ belongs to the inclusive graph precisely when $bq_{ij} \ge aA$, and to the strict graph when this inequality is strict. These comparisons govern discrete deletion and old-point witness construction. Floating thresholds used in insertion are only proposal mechanisms; their witness bounds are reassessed on decimal coordinates.

2.13 Non-degrading vertex deletion

The deletion routine seeks a smaller configuration whose lower bound is at least the preceding partition upper bound. It solves a normalized subset problem, allowing the diameter to shrink; it is not a search for a minimum-size subset at a fixed diameter.

Let S be the currently retained vertex indices and let τ be the current squared partition upper bound. Initially $S = [m]$ and $\tau = \upsilon$. For a proposed deletion $S' = S \setminus \{v\}$, recompute $A' = \max_{i,j \in S'} q_{ij}$ and form

$$E' = \{\{i, j\} \subset S' : q_{ij}/A' \geq \tau\}.$$

Only a definite UNSAT answer accepts the deletion. By Equation (1), it proves $R_k(X_{S'})^2 \geq \tau$. After acceptance, the old partition is restricted to S' and the next target becomes

$$\tau \leftarrow \frac{\max_{i,j \in S' :\, c(i)=c(j)} q_{ij}}{A'}.$$

Updating this target prevents a later deletion from losing a gain caused by the preceding diameter reduction. Both SAT and UNKNOWN leave the vertex in place.

Candidate vertices are ordered once by initial witness degree plus independent uniform noise, favoring low-degree vertices. Each vertex is tested at most once in that call, subject to the deletion budget and a retained count of at least $\max(k + 1, m_{\min})$. The default budget is 150 tests. Removed vertices remain isolated in the original oracle during these queries, avoiding a solver rebuild for every trial. At completion, rows and witness endpoints are compacted, the restricted partition is retained, and the resulting geometry is assessed again. The algorithm records deleted original indices. Its guarantee is non-degradation relative to the preceding upper bound; neither equality of the actual ratios nor minimum cardinality follows.

2.14 Alternative witnesses and exploratory compaction

Each tuning cycle can construct alternative witnesses on the current points. For a selected relative slack s, it includes all edges satisfying

$$q_{ij}/A \geq \lambda(1 - s)^2.$$

The implemented factor is rationalized from its computed decimal string, so subsequent edge comparisons use exact integers. The graph contains the known witness, and the baseline pruning routine removes edges only while preserving established non-colorability. Default slacks are $0, 0.004, 0.012, 0.03, 0.08, 0.15$; one old-point witness trial is proposed per cycle, with up to 80 edge-deletion tests and pruning noise 0.025.

Optional witness compaction retains all vertices incident to witness edges and enough unused vertices to respect the requested minimum count. Witness lengths are unchanged and the subset diameter cannot increase, so the witness lower bound cannot decrease. This is weaker than the deletion guarantee

in Section 2.13: preserving one witness bound does not establish preservation of the full configuration's unknown objective. Compacted configurations are exploratory proposals and must pass the incumbent comparison independently.

2.15 Tuner SLSQP and generation of diameter constraints

The tuner uses the fixed-witness problem in Equation (4), with two implementation differences from baseline polishing. First, it normalizes to diameter one and fixes $x_1 = 0$, eliminating translation. The variables are the remaining $(m-1)d$ coordinates and t. Coordinate bounds are $[-1.1, 1.1]$ and $0 \le t \le 1$; every actual diameter-one configuration with $x_1 = 0$ fits within these coordinate bounds.

Second, the default constraint set includes all witness constraints but only diameter constraints whose initial squared distances are at least 0.97, together with every pair incident to vertex 1. After each SLSQP solve, all pairwise distances at its terminal iterate are checked. Omitted constraints violating $\|x_i - x_j\|^2 \le 1$ by more than the feasibility tolerance are added, and the next solve starts from the normalized best retained geometry. The default limits are eight constraint-generation passes and 300 SLSQP iterations per pass. A full-constraint option includes every diameter inequality immediately.

Analytic Jacobians are supplied. Independently of the active constraint set, every retained callback or terminal candidate is scored with the true maximum over *all* pair distances. The initial geometry is also retained as a fallback. A useful intermediate geometry therefore cannot obtain an inflated ratio merely by violating an omitted diameter constraint. The terminal-success diagnostic requires solver success, full numerical feasibility, no omitted violated diameter constraints, and agreement of the terminal and retained witness ratios within the ratio tolerance. This last test compares ratios, not coordinate arrays. Checking that no omitted constraint is violated does not by itself mean that the included constraints are feasible or that the solver converged.

2.16 First-order LP audit and repeated polishing

At a normalized realization of H, write $s_e = \|x_i - x_j\|^2$ and $t_0 = \min_{e \in E(H)} s_e$. Define the near-active sets

$$\mathcal{D} = \{e : s_e \ge \max_f s_f - \epsilon_a\}, \qquad \mathcal{W} = \{e \in E(H) : s_e \le t_0 + \epsilon_a\}.$$

For coordinate velocities $v = (v_1, \dots, v_m)$ with $v_1 = 0$, let $\dot{s}_{ij}(v) = 2(x_i - x_j)^\mathsf{T}(v_i - v_j)$. The audit solves

$$\begin{aligned} \text{maximize} \quad & \eta, \\ \text{subject to} \quad & \dot{s}_e(v) \le 0 && (e \in \mathcal{D}), \\ & \eta \le \dot{s}_e(v) && (e \in \mathcal{W}), \\ & \|v\|_\infty \le 1, \quad 0 \le \eta \le 1. \end{aligned} \tag{5}$$

The default active tolerance is $\epsilon_a = 10^{-8}$. HiGHS solves this linear program with a 30-second time limit and primal and dual feasibility tolerances 10^{-9}. Near-active constraints deliberately restrict the available directions. Consequently, a small numerical optimum is only a diagnostic for this witness and tolerance, not an exact stationarity certificate or an exclusion of improvements using other witnesses.

Fixed-witness polishing repeats SLSQP and the LP audit, retaining only increases in the actual witness ratio. A pass counts as stagnant only when its gain is at most $\epsilon_r = 10^{-10}$ and its terminal-success diagnostic holds. After the required number of consecutive such passes (default two), the routine also tries the configured number of small Gaussian kicks (default one, coordinate standard deviation 10^{-4}), each followed by SLSQP. A kick resets the stagnant-pass counter, even when it does not improve the retained geometry.

If the LP gives a direction with gain exceeding 10^{-7}, it can be used as a proposal for nonlinear steps of sizes

$$10^{-2},\ 3 \cdot 10^{-3},\ 10^{-3},\ 10^{-4}, \ldots, 10^{-8}.$$

The first step increasing the actual witness ratio by more than ϵ_r is retained and normalized. The linear gain alone does not authorize acceptance. The status `numerically_stationary` requires enough successful stagnant passes, exhaustion of the configured kicks, and a successful audit with gain at most 10^{-7}. A final audit can downgrade this status. Exhausting the default eight polishing passes instead reports a budget limit. Numerical SLSQP tolerances (default objective tolerance 10^{-12} and feasibility tolerance 10^{-9}) and these reporting thresholds never replace exact incumbent comparisons.

2.17 Local tuning with witness refresh

A local trial starts from a known non-colorable graph. Optional smoothing uses the baseline loss of Equation (3), now with powers $32, 128, 512, 2048$ and a default 100 L-BFGS iterations per stage. The original decimal state is independently assessed and retained as the initial discrete fallback.

Each refresh applies repeated fixed-witness polishing, normalizes its output, rounds it into a new decimal geometry, and evaluates that geometry using the known witness and a retained partition. The local discrete state is replaced only if the new rational squared lower bound is at least its preceding squared upper bound. Thus a numerical improvement that fails after decimal rounding need not be retained.

Full evaluation may discover a stronger obstruction than the graph just polished. If the new lower endpoint exceeds the polished witness ratio by more than the ratio tolerance, the routine restarts from the retained state with its newly evaluated witness. It also refreshes when the candidate cannot support the retained-state reporting conditions. Otherwise it audits the retained decimal geometry and its retained witness before attaching a polishing report. A numerically stationary report may be transferred between numerically equivalent

ratios within the ratio tolerance, but is checked again by this retained-coordinate audit. This remains a numerical diagnostic. Failure of that audit downgrades the status; exhausting the default three refreshes gives a distinct refresh-budget status.

2.18 Insertion geometry and color-coverage refinement

Insertion proposes a block of new points with the aim of obtaining a new non-colorable witness. To diversify graph constructions, a small bank of old partitions is assembled. Starting with the saved partition, the program solves the strict graph above its upper-bound threshold under random vertex permutations, maps each coloring back, and deduplicates partitions up to color relabeling. The default target is three partitions; there is no enumeration guarantee.

Geometric candidates are sampled from the diameter-completion body

$$\mathcal{C}(X) = \bigcap_{i=1}^{m} \overline{B}(x_i, D),$$

which can extend beyond the convex hull. If c is the centroid, $z_i = x_i - c$, and u is a unit direction, the positive boundary parameter of the ray $c + ru$ is

$$r_*(u) = \min_i \left[u^\mathsf{T} z_i + \sqrt{(u^\mathsf{T} z_i)^2 + D^2 - \|z_i\|^2} \right]. \tag{6}$$

Directions mix normalized Gaussian vectors, centroid-to-vertex directions, and sums or differences of pairs of centered vertices. Most samples lie just inside r_*; independently selected interior samples (probability 0.22) use factors uniform on $[0.65, 0.98]$. With positive outer slack ω, a fraction of the remaining samples lie beyond the boundary, using factors $1 + \omega a$, $a \sim \mathcal{U}[0.15, 1]$. The default $\omega = 0.04$ therefore explicitly permits exploratory diameter expansion. Setting $\omega = 0$ removes this outer-shell proposal mechanism.

For a fixed old partition c, the color-coverage score of a proposed point y is

$$h_c(y) = \min_{a=1,\dots,k} \max_{i:\, c(i)=a} \|y - x_i\|^2.$$

A high score means that y has a distant neighbor in every color class. An empty class gives score $-\infty$. Promising candidates are refined by SLSQP inside $\mathcal{C}(X)$. For nonempty classes C_a, the smooth per-class values and objective are

$$M_{a,\beta}(y) = \frac{1}{\beta} \log \left(\frac{1}{|C_a|} \sum_{i \in C_a} e^{\beta \|y - x_i\|^2 / D^2} \right), \qquad f_\beta(y) = \frac{1}{\beta} \log \sum_{a=1}^{k} e^{-\beta M_{a,\beta}(y)}.$$

Minimizing f_β approximates maximizing coverage, using $\beta = 80,400$, analytic gradients, and a default 70 iterations per stage. Terminal points outside the completion body are retracted along the ray from the centroid by bisection. Retention compares the unsmoothed coverage score with that of the initial proposal. In particular, if the initial proposal came from the outer shell, the routine

can retain that outside point when feasible refinements do not improve its score; the later block diameter check is therefore still necessary.

The default pool contains 384 raw candidates, with up to 16 refinements. Points too close to an old vertex are filtered using a default separation of 10^{-5} in old-diameter units. At each comparison threshold, only one representative of each old-neighbor signature is kept. Blocks are sampled without replacement, with default size between one and three (the implementation permits at most six). New–new separations are checked as well. Every block must satisfy

$$\max_{i<j \text{ in the full block}} \|x_i - x_j\|^2 \le (1+\omega)^2 + 10^{-12}$$

in normalized coordinates. Testing only new–old distances would not suffice for this condition. These are floating proposal checks; decimal rounding, possible compaction, and subsequent free optimization require fresh evaluation and preclude treating every proposal as an exact ratio-preserving insertion.

2.19 Colorable skeletons and essential non-colorable extensions

For a partition from the bank and slack s, the comparison distance is $a = U_{\mathrm{src}}(1-s)$, where U_{src} denotes the source state's upper endpoint (the same endpoint is used for every bank member). The program builds a strict geometric graph at this distance on the old points and the proposed block. At zero slack, its squared cutoff is multiplied by $1 + 2 \cdot 10^{-11}$ as a numerical separation from tied bottleneck distances. This gap affects proposal selection only; the eventual bound uses actual decimal witness lengths.

From this full geometric graph F, remove all old–old edges whose endpoints have the same saved color, while keeping every edge incident to a new point. Denote the resulting comparison graph by J. Its old induced subgraph is colorable by construction. This step is essential: simply adding points to an already non-colorable old graph would not demonstrate any role for the insertion.

Before a full oracle call, the following screens reject proposals:

1. A new vertex has degree less than k in J.

2. A new vertex has the same neighbors outside a pair of vertices as another vertex in F (geometric false or true twins). False-twin folding has a color-extension justification when applied to the relevant graph; rejecting adjacent true twins is an exploration preference, since true twins can increase chromatic number. The geometric-twin screen is evaluated on F, not necessarily on J.

3. In J, a new vertex v is nonadjacent to some u and $N_J(v) \subseteq N_J(u)$. Such a vertex can reuse the color of u.

4. The particular old coloring extends to the entire new block.

For the last test, the available colors of each new vertex are those absent from its old neighbors. A small exact backtracking list-coloring search, ordered by increasing list size, checks the new–new edges. This handles cooperative blocks: individually available colors may be incompatible across the inserted vertices. Repeated comparison adjacency patterns are also skipped.

Failure to extend one coloring is only a screening condition. The full oracle must then establish that J is non-k-colorable, allowing all old vertices to recolor. Moreover, for *each* new vertex v, the oracle must report $J - v$ colorable. A non-colorable deletion indicates a redundant new point for this test, and an unknown deletion result leaves essentiality unestablished; both reject the proposal. Accepted blocks therefore satisfy

$$J[\text{old vertices}] \text{ is colorable}, \qquad J \text{ is non-colorable},$$
$$J - v \text{ is colorable for every new } v.$$

This establishes the relevance of each new vertex in the specified comparison graph. It is not a statement that deleting that vertex from the full geometric threshold graph restores colorability, or that the normalized finite-set objective has improved.

After this test, the witness or assumption core is pruned with definite UNSAT preservation and optionally compacted. Its bound is computed on the new decimal geometry. The default screening limits are 3000 cheap attempts and 150 full-extension tests per cycle; the latter counter excludes the additional per-vertex essentiality queries and pruning calls. Up to three times the requested number of successful proposals are collected, ranked by actual floating witness ratio with smaller point count as a tie-breaker, and truncated to the default six local insertion trials. Point-count limits also apply. These budgets and screens select proposals, not an exhaustive family of extensions.

2.20 Tuning cycles, incumbent acceptance, and saved results

The workflow first reassesses the input, attempts non-degrading deletion, and locally polishes the retained incumbent. Each subsequent cycle (default 20) chooses a source, reduces it, generates old-point alternative witnesses and insertion proposals, then assesses and locally tunes each proposal. A proposal may update the incumbent both before and after polishing. Trial outputs and proposal metadata are retained separately from the scientific incumbent.

An exploratory archive holds at most eight trial results, ranked by lower endpoint and then point count. Entries must lie within the relative lower-endpoint deficit $\epsilon = 0.002$ of the incumbent by default. Every third cycle may draw its source uniformly from this archive; other cycles use the incumbent. Archive membership uses floating scores and is not an incumbent acceptance test. Near-record exploratory branches can therefore support later moves without being reported as improvements.

Let $(\lambda_o, \upsilon_o, m_o)$ be the incumbent's rational squared endpoints and point count. A new geometry is saved when it gives a strict certified improvement

$\lambda_n > \upsilon_o$, or when $\lambda_n \geq \upsilon_o$ and $m_n < m_o$. An additional plateau rule allows $\lambda_n \geq \upsilon_o$, $m_n \leq m_o$, and a newly acquired `numerically_stationary` status. That status remains a numerical report, not an optimality theorem. At exactly the same stored rows and precision, a changed nonempty polishing report may also be saved if the bracket does not widen: $\lambda_n \geq \lambda_o$ and $\upsilon_n \leq \upsilon_o$. This permits report or bound refinement of an unresolved state without asserting a strict objective gain. These rules differ from baseline lower-endpoint ranking and from the local non-strict acceptance inside Section 2.17.

After all cycles, the incumbent itself receives a final deletion attempt and local tuning without smoothing, including the retained-coordinate LP audit. A polish-only mode disables cycles, deletions, and smoothing but still uses the local polishing and assessment machinery. Exhausted polishing or refresh budgets remain visible in the output status.

Every accepted checkpoint is written to a numbered history directory; a staged copy replaces the current best while preserving its predecessor. The coordinate and certificate format is compatible with baseline verification, with the geometry's actual decimal precision retained and an additional file containing polishing reports. Integer geometry produces outward-rounded decimal bounds as in Section 2.10. The tuner does not independently proof-check SAT solver UNSAT results, and an LP audit of the floating image of saved decimal coordinates is still numerical. Neither a smaller witness, an essential insertion, nor a numerical stationary status establishes global extremality.

2.21 Single-vertex splitting: purpose and initialization

The program `split_tune.py` replaces one vertex by two nonadjacent copies with different incident witness constraints. The copies initially occupy the same position, so each may subsequently move under a different set of lower-distance constraints. This increases the number of coordinate variables without initially reducing the chosen witness bound. The method uses the tuner's decimal state, integer evaluator, SLSQP implementation, and checkpoint machinery, with a separate graph construction and screening procedure. It supports general $k \geq 2$, with $k = d + 1$ when no saved or explicit value is supplied.

Initialization loads and reassesses the input as in Section 2.11. Unless reductions are disabled, it applies the exact non-degrading vertex-deletion routine with a budget equal to the initial number of points and a minimum retained count of $k + 1$. There is no initial continuous polishing stage before split discovery. The default point limit is the retained starting count plus the round budget (20 rounds by default). Each round constructs its proposals from the incumbent at the start of that round.

Let λ be that state's rational squared lower endpoint and let q_e and A denote its integer squared distances and diameter numerator. The initial discovery graph is

$$E(H) = \{e : q_e/A \geq \lambda\}.$$

It contains the verified witness and is therefore non-k-colorable, even if the state

has an unresolved bracket. Optional UNSAT-preserving pruning can reduce this graph before splitting; it is disabled by default. When enabled, pruning uses the requested edge-deletion budget, noise 0.04, and a random generator seeded by the global seed plus the point count. All subsequent edge-criticality statements refer to this particular possibly pruned graph H.

2.22 Near-shortest edges and omission-set enumeration

Normalize a floating copy of the source geometry to diameter one, and write s_e for its squared distances. Discovery restricts deletions to near-shortest witness edges

$$E_a = \{e \in E(H) : s_e \le \min_{f \in E(H)} s_f + \epsilon_a\}, \qquad \epsilon_a = 10^{-8} \text{ by default.}$$

For each such edge e, query $H - e$. Mark e critical only when this graph receives a definite SAT answer. A UNSAT answer shows that this edge is not necessary for the selected obstruction; UNKNOWN establishes neither conclusion. Criticality is a combinatorial property of H, whereas membership in E_a is a numerical proposal filter.

For a vertex v, let $N_a(v)$ be its neighbors along E_a and let $C(v) \subseteq N_a(v)$ correspond to the edges just proved critical. Enumerate omission sets $S \subseteq N_a(v)$ satisfying

$$1 \le |S| \le r, \qquad S \cap C(v) \neq \varnothing.$$

The default r is 2, and the implemented choices are 1, 2, or 3. Candidate splits use unordered pairs (L, R) of these sets with $L \cap R = \varnothing$. Exchanging L and R merely exchanges the two copies, so only one ordering is tested. In particular, a tested vertex must have at least two distinct critical near-shortest incident edges. This does not assert that every useful split must have that property.

If $a_v = |N_a(v)|$ and $c_v = |C(v)|$, the number of eligible individual sets is

$$M_v = \sum_{j=1}^{\min(r, a_v)} \left[\binom{a_v}{j} - \binom{a_v - c_v}{j} \right],$$

where an impossible binomial choice is zero. Before disjointness filtering, there are at most $\binom{M_v}{2}$ pairs. Thus even small omission sizes can lead to many graph queries. Vertices and subsets are traversed in index order. An optional screen limit stops after a specified number of split-graph queries; zero means no such limit. This counter excludes the preceding critical-edge queries, optional pruning, and LP audits. The tested family excludes longer-edge omissions, intersecting omissions, and omissions exceeding r; it is not an exhaustive search over graph surgeries. A shared omitted edge that is critical would make the split colorable by duplicating its endpoint in a coloring of $H - e$, but this argument does not justify a universal claim about every intersecting pair.

2.23 Split graph and essentiality of both copies

Retain vertex v as the first copy and append a new vertex v'. All edges not incident to v remain unchanged, and define

$$\begin{aligned} E(H_{v;L,R}) = E(H - v) \\ \cup \{\{v, u\} : u \in N_H(v) \setminus L\} \\ \cup \{\{v', u\} : u \in N_H(v) \setminus R\}. \end{aligned} \tag{7}$$

There is no edge $\{v, v'\}$. Disjoint nonempty omissions make the two neighborhoods different, and their union is the original neighborhood. The construction validates endpoints, remaps all pair indices to the new point count, and introduces no other edges. It is a graph operation; non-colorability is not automatic.

The omission conditions give a useful essentiality guarantee. Choose $\ell \in L \cap C(v)$ and $r_0 \in R \cap C(v)$. Deleting v' from $H_{v;L,R}$ leaves H with all v–L edges removed, a subgraph of the known colorable graph $H - \{v, \ell\}$. Deleting v instead leaves, after relabeling v' as v, a subgraph of the colorable graph $H - \{v, r_0\}$. Hence both single-copy deletions are colorable. If a separate oracle query establishes

$$H_{v;L,R} \text{ is non-}k\text{-colorable},$$

then both copies are essential to this obstruction. No additional copy-deletion queries are needed for that scaffold-level conclusion. SAT and UNKNOWN split answers are rejected. The conclusion concerns the selected graph, not every vertex of that graph, its minimum size, or the full geometric threshold graph after optimization.

For geometric initialization, set $x_{v'} = x_v$. All split edges then have lengths inherited from H, every old edge length remains represented because L and R are disjoint, and the diameter is unchanged. Thus

$$L_{H_{v;L,R}}(x_1, \ldots, x_m, x_v) = L_H(X).$$

The zero distance between the two copies is harmless for this witness because they are nonadjacent. Nevertheless, the coincident coordinate array is only an auxiliary graph realization, not a finite set of $m + 1$ distinct points and not an accepted decimal state.

2.24 LP screening and trial selection

Every split receiving UNSAT is audited at its coincident realization using the first-order LP in Equation (5). The audit includes the near-active diameter constraints of the enlarged array and the near-shortest edges of the split witness. An unsuccessful LP solve discards the proposal. Successful audits are retained with their numerical gains; a gain exceeding $\epsilon_{\mathrm{LP}} = 10^{-7}$ is called positive for selection purposes. The split discovery tolerance is also used as the LP near-active tolerance, with a default LP time limit of 30 seconds per audit.

Positive-gain splits are ranked in descending gain order. Defaults select at most 24 such trials per round and at most four for any original vertex,

to avoid spending the entire optimization budget on one vertex's variants. A zero value for either cap removes that cap. Optionally, a specified number of successful audits with gain at most ϵ_{LP} are sampled in a seeded random order and appended. These extra trials do not count toward the positive-gain or per-vertex caps. They are disabled by default. Their label "zero gain" includes small positive gains below the threshold and is not an exact assertion of zero derivative.

A large LP gain is only a ranking signal for the fixed split witness. Near-active constraints and numerical tolerances can affect the score, and a positive directional prediction need not yield an accepted nonlinear improvement. Conversely, a small score does not rule out finite-step, higher-order, or other-witness improvements. The LP direction itself is not used as the starting displacement of the default positive-gain trial: SLSQP starts directly from the coincident realization.

2.25 Separation by SLSQP and exact trial acceptance

For each selected split, the program reconstructs the normalized source coordinates and appends the coincident copy. For an optional zero-gain trial, it instead first separates the copies symmetrically:

$$x_v \leftarrow x_v - \zeta u, \qquad x_{v'} \leftarrow x_v^{\mathrm{old}} + \zeta u,$$

where u is a normalized Gaussian direction and the default $\zeta = 0.002$ is in source-diameter units. Both expressions use the original copy position as their center. The perturbation changes the geometry, so preservation of the initial witness ratio is no longer assumed.

A trial then calls the tuner's single SLSQP routine from Section 2.15, with a default 300 iterations and objective tolerance 10^{-12}. A single call may still include generated-constraint repetitions (eight passes by the inherited default). All coordinates except the translation anchor are free, so this is not optimization of just the new copy. As in the tuner, the routine retains the best actual nonlinear witness ratio, including the recomputed full diameter, independently of its terminal success flag. No L-BFGS smoothing or repeated polishing with kicks is used in this trial stage.

The returned coordinates are normalized, rounded to 17 decimal places, and passed to the integer evaluator with the split graph already known non-colorable. Duplicate rounded rows reject the trial; in particular, an unseparated coincident pair is never admitted as an enlarged decimal set. The complete objective can be stronger than the split-witness bound, since reassessment may identify another obstruction.

Writing (λ_*, υ_*) for the *current* incumbent's rational squared bracket, a split trial is accepted only if both

$$\sqrt{\lambda_{\mathrm{trial}}} > \sqrt{\lambda_*} + \gamma, \qquad \lambda_{\mathrm{trial}} > \upsilon_* \tag{8}$$

hold. The first condition is evaluated numerically with default minimum gain $\gamma = 10^{-10}$; the second is an exact rational comparison. An unresolved trial can

therefore be accepted only when its established lower bound already exceeds the incumbent upper bound. Neither the LP gain, SLSQP objective, nor solver success can substitute for these tests. This strict rule applies to split proposals; inherited reductions and final polishing use the tuner's own non-degrading and report-refinement policies.

Every selected trial in a round derives from the same round-start source, even after an earlier trial is accepted. Acceptance compares against the latest incumbent, and the round continues through the remaining selected trials. Consequently, several incumbent updates within one round are alternative splits of a common source, not successive splits of one another. Successive graph surgeries occur only when a later round starts from the newly retained configuration.

2.26 Diagnostics on the accepted geometric threshold

Before publishing an accepted trial, the program constructs the inclusive threshold graph of its actual decimal coordinates at its rational squared lower endpoint. This graph can contain edges absent from the split scaffold. It compares the original and appended vertices' neighborhoods outside that pair, records their floating Euclidean separation, and queries the threshold graph after deleting each copy's incident edges. The diagnostic is therefore about the accepted geometry, rather than merely reusing the scaffold essentiality argument.

A SAT deletion means that this particular fixed-threshold graph becomes colorable when the indicated vertex is removed; UNSAT means a non-colorable obstruction remains, and UNKNOWN is unresolved. The queries keep the deleted vertex isolated and do not renormalize a geometric subset. Their answers alone thus give no non-degradation statement for the subset's normalized ratio. The reported twin flag compares neighborhoods outside the pair and does not distinguish adjacent true twins from nonadjacent false twins.

These diagnostics are descriptive, not extra acceptance gates. A trial passing Equation (8) can be saved even if these new threshold tests do not show both copies essential. Distinctness is checked for all decimal rows, but there is no positive minimum-separation acceptance threshold. Only the selected pair's separation is recorded; near coincidences with other points are not globally audited here. The split diagnostics describe the accepted trial before any subsequent vertex reduction or final polishing, which can change indices, distances, and threshold adjacencies. Such later states require their own diagnostics if those properties are to be claimed for the final output.

2.27 Split rounds, termination, and final polishing

The complete one-step search can be summarized as follows:

1. Reassess the input and optionally reduce its point count using exact non-degrading deletion.

2. Construct the current threshold witness, optionally prune it, and test its near-shortest edges for criticality.

3. Enumerate disjoint omission pairs. Retain split graphs only after definite non-colorability and a successful LP audit.

4. Select trials by numerical LP gain and the vertex-diversity caps; optionally include a sample of small-gain trials.

5. Optimize each selected scaffold, reject coincident decimal outputs, reassess its bounds, and apply the strict split acceptance test.

6. If the round improves the incumbent, optionally reduce it again and begin the next round; otherwise terminate the split search.

7. Apply a separate final polishing and audit procedure to the incumbent.

The recorded stopping reason distinguishes exhaustion of the round or point budget, absence of selected splits in the tested family, and failure to improve among the selected splits. A round counts as accepted if at least one of its proposals improves the incumbent. Reduction after an accepted round may remove old points or copies, so the net point-count increase need not be one. Neutral scaffolds are not carried forward to enable a second split in this workflow. Failure of the one-step search therefore does not exclude a productive sequence starting with a neutral split.

Final polishing reuses `local_tune` with smoothing disabled, all diameter constraints included, up to three witness refreshes, the requested polishing-pass budget (default eight), and one numerical kick. Its LP stationarity threshold is the split screening threshold. The final candidate is considered by the tuner's checkpoint policy, which can retain a non-degrading simplification or numerical-report refinement as well as a strict gain. Attempting this polish does not guarantee its convergence or replacement of the incumbent; the retained status is reported separately from the split-search stopping reason.

Per-round output records screening counts, selected omission sets and LP audits. Accepted-trial metadata additionally records SLSQP results, bounds, and geometric diagnostics. Optional trial saving stores every successfully evaluated trial, including those not accepted. Numbered checkpoint histories and final certificates use the tuner's decimal format. The non-colorability trust boundary and numerical-audit limitations remain those of Sections 2.10 and 2.16: no independent SAT proof checking, global optimality claim, or exact local-optimality certificate is supplied.

2.28 Paired splitting: scope and initial graph

The canonical module `paired_splits.py` explores two successive essential vertex splits before optimizing their common realization. This addresses a limitation of the one-step workflow: a first split that does not immediately improve the geometry may change the constraints so that a second split permits useful motion. The first move is retained as an auxiliary graph, without requiring a positive LP gain or an improvement of the finite-set objective. The program

does not actually classify first moves by their individual LP gains; it keeps all tested first splits established non-colorable, subject to branch selection.

The current implementation fixes $k = 4$ throughout the oracle and export paths. Saved inputs declaring another k are rejected before an output directory is created; inputs without such metadata are interpreted with $k = 4$. No corresponding restriction fixes the coordinate dimension to three, but the program is not a general-k paired-search interface. It reuses the tuner's exact-decimal loading and assessment. The initial state is saved separately, and its inclusive integer threshold at the lower endpoint supplies the source graph H_0. Optional witness pruning uses the baseline UNSAT-preserving routine with noise 0.04 and is disabled by default. There is no initial vertex reduction or polishing. A floating copy $X = (x_1, \dots, x_m)$ is normalized to diameter one.

All paired proposals derive from this fixed source geometry and graph. An accepted improvement updates the comparison incumbent but does not restart graph enumeration from the improved coordinates. Thus one invocation explores a family of two-split proposals, rather than a chain of accepted split rounds or a recursive search tree of arbitrary depth.

2.29 Enumeration of two successive essential splits

At either stage, the internal split enumerator applies the construction of Sections 2.22 and 2.23. It selects edges within 10^{-8} in squared distance of the current witness's shortest edge, tests their individual deletions, and marks critical edges only after a definite SAT answer. Each of the two disjoint, nonempty omission sets must contain a critical neighbor. The default maximum omission size is three. Unlike the one-step command-line interface, the paired interface does not restrict this parameter to the choices 1, 2, and 3; larger supplied values expand the family up to the available neighborhood size.

For a first split at vertex v, denote its omission sets by L_1, R_1 and construct

$$H_1 = (H_0)_{v;L_1,R_1}, \qquad Y = (x_1, \dots, x_m, x_v).$$

Only definite UNSAT first splits are retained. The entire first family is enumerated before the retained list is shuffled using the seeded random generator. An optional first-branch limit then truncates that shuffled list. It limits the number of branches explored further, not the number of first-stage oracle calls or enumeration time. By default this limit is zero, meaning no truncation.

For each retained first branch, the enumerator recomputes near-shortest and critical edges on H_1 at the still-coincident realization Y. The second split acts on an original vertex w with $v < w \le m$; the newly appended copy v' is excluded as a split target. Its omission sets L_2, R_2 are chosen from the neighbors in H_1, so they may involve the first copy even though that copy cannot itself be split. Define

$$H_2 = (H_1)_{w;L_2,R_2}, \qquad Z = (x_1, \dots, x_m, x_v, x_w). \tag{9}$$

The second graph uses the enlarged pair indexing and is independently queried for non-colorability. SAT and UNKNOWN answers are counted but discarded.

In particular, the second graph is not formed by independently splitting two neighborhoods in H_0: its edges and criticality tests depend on the first transformation.

The restriction $v < w$ chooses one ordering of the original split targets; it should not be interpreted as a proved exhaustive symmetry reduction for every possible ordered surgery. The implementation also excludes resplitting the first vertex or its appended copy. A global pair-test limit, zero by default, counts all second-stage split queries regardless of their answer. It excludes first-family enumeration, critical-edge queries, LP audits, and optimization. Criticality screening of a visited branch occurs before its second-stage split queries are yielded. These limits therefore do not directly bound total computational time.

2.30 Essentiality and coincident paired realizations

Each accepted stage has the copy-essentiality guarantee proved in Section 2.23. For the second pair, deleting either w or w' leaves a colorable subgraph of a critical-edge deletion of H_1. The first pair's essentiality also persists in the final non-colorable scaffold. To see this, delete v or v' from H_1; that graph is colorable. Applying the second split to a colorable graph preserves colorability: assign the new copy w' the color of w, since they are nonadjacent and each has only a subset of the former neighbors of w. Removing either member of the first pair from H_2 gives just such a split of a colorable graph. Consequently, conditional on the established non-colorability of H_2,

$$H_2 - u \text{ is 4-colorable for each } u \in \{v, v', w, w'\}.$$

This argument concerns the constructed scaffold; the implementation does not separately query these four deletions at the final optimized geometric threshold. It also does not establish criticality of every other vertex or minimum graph size.

Both pairs start at coincidence in Z. By the disjoint-omission construction, each stage preserves the represented old witness lengths and introduces no greater distance, so

$$L_{H_2}(Z) = L_{H_1}(Y) = L_{H_0}(X).$$

The two zero-length copy pairs are nonedges of the scaffold. The array therefore supplies a valid auxiliary realization for its witness constraints but not a distinct $(m+2)$-point configuration. The graph and continuous steps must eventually produce distinct decimal rows before an improvement can be exported. All later diameter calculations include every pair of coordinates, including pairs between copies of different original vertices.

2.31 Joint LP screening and small-gain sampling

Every second-stage graph receiving UNSAT is audited at Z using the tuner's LP of Equation (5), with its default near-active tolerance 10^{-8} and LP time limit of 30 seconds. There is no LP audit of H_1 as an admission condition. A failed joint

audit is discarded; a successful joint gain greater than 10^{-7} triggers immediate optimization of the paired scaffold. These positive candidates are processed in enumeration order, without the one-step module's per-vertex quota or sorting by LP gain.

Successful audits with gain at most 10^{-7} enter a bounded collection for later perturbed trials, with default capacity 24. The first such candidates fill the collection. Once full, the program draws a uniform integer $j \in \{0, \dots, S - 1\}$, where S is the cumulative number of UNSAT split answers counted so far, including the entire first family and positive as well as small-gain second splits. If j is less than the collection capacity, entry j is replaced by the current candidate. This is the implemented sampling rule; it is not uniform reservoir sampling over the small-gain population, whose count is not the denominator. Setting the capacity to zero disables these later trials without disabling enumeration or LP screening.

After paired enumeration stops, each retained small-gain seed receives independent Gaussian noise in *every* coordinate, with default standard deviation 0.01 in normalized source-diameter units, and the whole array is normalized again. This differs from the one-step splitter's opposite perturbation of only one pair. The resulting finite displacement can reveal motion missed by the near-active LP, but does not preserve the initial witness ratio by assumption. The small-gain label includes numerical zero and gains below the threshold; it is not an exact stationarity claim. Positive-gain seeds instead start SLSQP directly at coincidence, without stepping along the LP direction.

2.32 Joint SLSQP and publication of improvements

Both positive and perturbed small-gain trials call the tuner's single SLSQP routine on the full paired witness. It uses generated diameter constraints, 350 SLSQP iterations per solve, the inherited maximum of eight constraint-generation passes, and objective tolerance 10^{-12}. All coordinates other than the translation anchor can move. This stage retains the best actual witness ratio from the seed, callback iterates, and endpoint, recomputing the full diameter as in Section 2.15. Terminal solver success is recorded but is not a publication condition.

Let r_H be the floating witness ratio of the returned geometry and (λ_*, υ_*) the current incumbent's rational squared bracket. The paired workflow first applies the cheap numerical filter

$$r_H > \sqrt{\lambda_*} + 10^{-10}.$$

Only candidates passing this filter are normalized, rounded to 17 places, and fully assessed by the integer evaluator with H_2 known non-colorable. Invalid geometry, including coincident decimal rows, is rejected. An assessed candidate replaces the incumbent only if

$$\lambda_{\text{trial}} > \upsilon_*. \tag{10}$$

Thus publication requires a strict established improvement on the saved rational geometry. An unresolved bracket can qualify if this inequality holds; solver timeout alone cannot make it hold.

The prefilter uses the optimized *fixed-witness* ratio rather than the lower endpoint of a full evaluation. It may therefore skip a geometry whose stronger obstruction would have been discovered by reassessment. There is no separate numerical minimum-gain test on the final integer state, and no requirement that it acquire a stationary-polish status. These details distinguish paired acceptance from both the one-step split acceptance rule and the tuner's checkpoint/report refinements. The baseline solver, numerical LP, and exact-coordinate trust boundaries otherwise remain unchanged.

Every successful paired proposal contains exactly $m + 2$ decimal points; this module performs no post-trial deletion or witness compaction. Distinctness is enforced, but no global minimum-separation threshold is checked or reported. Unlike the one-step module, it does not invoke split-specific twin or copy-deletion diagnostics on an accepted geometry. Scaffold essentiality must therefore not be asserted for its full optimized threshold graph without an additional audit.

2.33 Paired-search outputs and termination

The program can be summarized by the following sequence:

1. Reassess the input with $k = 4$, save it, and construct or optionally prune its inclusive lower-bound threshold graph.

2. Enumerate all non-colorable first splits, shuffle them, and apply the optional retained-branch limit.

3. For each retained branch, enumerate second splits of later original vertices and retain only definite non-colorable scaffolds.

4. Audit the joint coincident realization. Immediately optimize positive-gain scaffolds and maintain the bounded small-gain collection.

5. After enumeration or its pair-test limit, perturb and optimize the retained small-gain scaffolds.

6. Publish only strict exact-bound improvements and write the summary.

There is no final repeated full-constraint polish, witness-refresh loop, or final LP audit of the saved coordinates. Such post-processing belongs to a separate tuner invocation. The LP metadata saved with a proposal concerns the initial coincident scaffold, even when the geometry later undergoes jitter and nonlinear optimization.

The input snapshot is always saved separately. Each improvement receives its own numbered directory and updates the best directory using the tuner's state-export routine. The improvement directory additionally stores the actual optimization seed and the fixed optimization edges; these can differ from the

witness returned by full assessment. The ordinary certificate and its authoritative decimal CSV concern the optimized distinct set, while a saved auxiliary NumPy seed may contain coincident rows. SLSQP and initial LP reports are proposal metadata; no repeated-polishing report is attached to the state, whose separate polish file can therefore say that it was not polished despite the trial SLSQP call.

If no trial satisfies Equation (10), the summary reports the input as the retained best state and no best directory is created. Progress records count first and second split answers, LP-positive pairs, optimization calls, and oracle statistics. Most optimized trials are logged even when not accepted, but geometry-conversion failures return before the usual optimized-trial event and do not produce a dedicated rejection entry. The summary does not contain a separate search-stop reason, and the module does not write a complete configuration file. A reproducible experiment report should therefore retain the exact invocation, input and source hashes, and the external run budget.

The procedure stops on exhaustion of the retained two-split family or its optional pair-test limit, then finishes the retained perturbed trials. Neither outcome excludes useful omitted branches, other split orders, longer split sequences, or geometries skipped by numerical screening. All first-stage and paired neutral states are exploratory; only the integer-geometry comparison establishes a published improvement, and no global or exact local optimality claim follows.